

```
[6]: import pandas as pd
a=pd.series([1,2,3,4,5])
print(a)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[6], line 2
      1 import pandas as pd
----> 2 a=pd.series([1,2,3,4,5])
      3 print(a)

AttributeError: module 'pandas' has no attribute 'series'
```

```
[2]: !pip install pandas
```

```
Collecting pandas
  Downloading pandas-3.0.3-cp314-cp314-win_amd64.whl.metadata (19 kB)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\KPRCAS\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\KPRCAS\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\KPRCAS\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in C:\Users\KPRCAS\AppData\Local\Programs\Python\Python314\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading pandas-3.0.3-cp314-cp314-win_amd64.whl (9.9 MB)
----- 0.0/9.9 MB ? eta -:-:--
----- 1.6/9.9 MB 12.9 MB/s eta 0:00:01
----- 4.2/9.9 MB 12.8 MB/s eta 0:00:01
----- 6.6/9.9 MB 12.2 MB/s eta 0:00:01
----- 8.4/9.9 MB 11.2 MB/s eta 0:00:01
----- 9.7/9.9 MB 10.4 MB/s eta 0:00:01
----- 9.9/9.9 MB 9.9 MB/s 0:00:01

Installing collected packages: pandas
Successfully installed pandas-3.0.3
```

```
[5]: import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)
```

```
[5]: import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

```
[11]: import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)
print(a.dtype)
print(a.index)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
int64
RangeIndex(start=0, stop=5, step=1)
```

```
[12]: import pandas as pd
a=pd.Series([True,False,True])
print(a)
print(a.dtype)
print(a.index)
```

```
0    True
1   False
2    True
dtype: bool
bool
RangeIndex(start=0, stop=3, step=1)
```

```

1
2 ram
3 dhakshi
4 amritha
5 jocima
dtype: str
str
RangeIndex(start=0, stop=6, step=1)
alice
alice
alice
1 alice
2 ram
dtype: str
1 alice
2 ram
3 dhakshi
dtype: str
2 ram
3 dhakshi
4 amritha
dtype: str

```

```

[52]: import pandas as pd
arr=pd.Series([31,25,15,33,42,15],index=['A','B','C','D','E','F'])
print(arr)
print(arr['E'])
#vectorized operation-no loop required
print(arr+5)
print(arr.iloc[1])
print(arr.loc['A'])
# statistical function
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())
#print(arr.percentile(arr,25)) is not available

```

```
C 20
D 38
E 47
F 20
dtype: int64
25
31
161
10.666145820617054
42
15
26.833333333333332
28.0
A 31
D 33
E 42
dtype: int64
15 2
31 1
25 1
33 1
42 1
Name: count, dtype: int64
0 PEN
1 PENCIL
2 BOOK
3 NOTE
4 PAPER
dtype: str
0 3
1 6
2 4
3 4
4 5
dtype: int64
```

[55]:

```
import pandas as pd
a=pd.Series([
    "gayathri":12,
    "gayathri":22,
```

```
arr=pd.Series([31,25,15,33,42,15],index=['A','B','C','D','E','F'])
print(arr)
print(arr['E'])
#vectorized operation-no loop required
print(arr+5)
print(arr.iloc[1])
print(arr.loc['A'])
# statistical function
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())
#print(arr.percentile(arr,25)) is not available
#boolean indexing
print(arr[arr>30])
# it return individual value count
print(arr.value_counts())
#string operation
name=pd.Series(['pen','pencil','book','note','paper'])
print(name.str.upper())
print(name.str.len())
```

```
A    31
B    25
C    15
D    33
E    42
F    15
dtype: int64
42
A    36
B    30
C    20
D    38
E    47
F    20
```

```
print(arr[['A','C','F']])
```

```
A    31
C    15
F    15
dtype: int64
```

```
[108]: import pandas as pd
#method1 : from dictionary
data={
    'name':['monika','nagul','perm'],
    'rollno':[27,20,24],
    'age':[19,20,21]}
df=pd.DataFrame(data)
print(df)

s1=pd.Series([1,2,3,4,5])
s2=pd.Series([5,6,9.54,9.4,54.60,6.00])
s3=pd.Series([True,False,True,False])
df=pd.DataFrame({'sno':s1,
                 'cgpa':s2,
                 'attendance':s3})

print(df)
#list of list
data=[['athish',22,98],['rithika',20,89],['prathiksha',24,21]]
df=pd.DataFrame(data,columns=["name","age","mark"])
print(df)

#list of dictionary
data=[{"name":"sakthi","course":"ds","grade":"a"},
      {"name":"madhu","course":"aiml","grade":"a+"},
      {"name":"ragavi","course":"csda","grade":"b"}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df)

#adding one extra columns
df["marks"]=[98,99,97]
```

```
#list of dictionary
data=[{"name":"sakthi","course":"ds","grade":"a"},
      {"name":"madhu","course":"aiml","grade":"a+"},
      {"name":"ragavi","course":"csda","grade":"b"}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df)

#adding one extra columns
df["marks"]=[98,99,97]
print(df)

#adding one row
std4=pd.DataFrame([{"name":"mahi","course":"da","grade":"o","marks":87}],index=["std4"])
df=pd.concat([df,std4])
print(df)

#create multiple rows
students=pd.DataFrame([{"name":"sanjiv","course":"csda","grade":"a","marks":97},
                       {"name":"divya","course":"da","grade":"o","marks":97}],index=["std5","students"])
df=pd.concat([df,students])
print(df)

#column access by brackets notation and dot notation
print(df.course)
print(df[['grade','marks']])

print(df.shape)
#rows
print(df.shape[0])
#columns
print(df.shape[1])
#dtype
print(df.dtypes)
#info
print(df.info())
print(df.describe())
#select specific row
print(df.loc[['std1','std4']])
print(df.iloc[3])
#update value
```

```
#delete column
df.drop('grade',axis=1,inplace=True)
print(df)
#delete row
df.drop('students',axis=0,inplace=True)
print(df)
#sorting
df.sort_values('marks',inplace=True)
print(df)
df.sort_values('marks',ascending=False,inplace=True)
print(df)
#statistical function for particular column
print(df['marks'].max())
```

```

      name  rollno  age
0  monika      27   19
1   nagul      20   20
2   perm      24   21
   sno  cgpa  attendance
0  1.0   5.00         True
1  2.0   6.00        False
2  3.0   9.54         True
3  4.0   9.40        False
4  5.0  54.60         NaN
5  NaN   6.00         NaN
      name  age  mark
0   athish   22   98
1   rithika   20   89
2  prathiksha  24   21
      name  course  grade
std1  sakthi     ds     a
std2   madhu    aiml     a
```

```
[113]: import pandas as pd
#read csv file
df=pd.read_csv('customers-100.csv')
print(df)
#read only 1st 5 data
```




```
std4      da
std5      csda
students  da
```

```
[113]: import pandas as pd
#read csv file
df=pd.read_csv('customers-100.csv')
print(df)
#read only 1st 5 data
print(df.head())
print(df.tail())

print(df['sex'])
```

```
3  Dominguez, Mcmillan and Donovan      Bensonview
4      Martin, Lang and Andrade      West Priscilla
..
95      Carey LLC      East Richard
96      Ware, Burns and Oneal      New Bradview
97      Schmitt-Jones      Ortegaland
98      Fitzgerald-Harrell      Lake Elijahview
99      Le, Nash and Cross      Judymouth

Country      Phone 1 \
0      Chile      229.077.5154
1      Djibouti      5153435776
2      Antigua and Barbuda      +1-539-402-0259
3      Dominican Republic      001-808-617-6467x12895
4      Slovakia (Slovak Republic)      001-234-203-0635x76146
..
95      Guyana      (188)169-1674x58692
96      Sri Lanka      +1-846-706-2218
97      Solomon Islands      +1-753-067-8419x2170
```

[]: